

GADING ADITYA PERDANA

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Central Jakarta, Indonesia

SUMMARY

First-generation undergraduate AI Researcher and Engineer specializing in Computer Vision and Deep Learning. Author of five peer-reviewed publications (four first-author) with interest in Vision Transformers, Ensemble Learning, and Model Calibration. Independently drive research projects from conception to publication while self-funding through freelance work. Experienced in implementing and deploying machine learning systems, with strong foundations in PyTorch.

EDUCATION

Binus University | GPA: 3.55/4.00

Jakarta, Indonesia

Bachelor of Science in Computer Science, Specialization in Intelligence Systems

Aug. 2023 – Feb 2027

- Accelerated 3.5-year program; **Apple Developer Academy Scholar (Cohort 2026)**
- Research focus: Medical Imaging, Vision Transformers, and Efficient AI.

PUBLICATIONS

- **Perdana, G. A.**, Nasari, M., Saputra, M.A., Halim, R., Minor, K.A. "Enhancing Bone Fracture Detection in X-ray Images Using Faster R-CNN with EfficientNetV2." *IEEE ISRITI* (2025). **First author.**
- **Perdana, G. A.**, Ghazali, M.A., Iswanto, I.A., Joddy, S. "CALM: Calibrated Adaptive Learning via Mutual-Ensemble Fusion." *Procedia Computer Science* (ICCCSI 2025). **First author. [1 Citation]**
- **Perdana, G. A.**, Kisdi, M.I.A., Wairooy, I.K., Makalew, B.A. "Analytical Analysis of Cryptocurrency Regulation and Adoption: A Machine Learning-Driven Ablation Study of the United States, Russia, and Indonesia." *Procedia Computer Science* (ICCCSI 2025). **First author. [3 Citations]**
- Nathen, A., **Perdana, G. A.**, Jefferson, G., Hasani, M.F., Maulina, A., Tjahyadi, B.G. "Tiny vs. Tinier: Baseline ViT-Tiny vs. Ensemble-Distilled Student on Imbalanced Fracture Detection." *Procedia Computer Science* (ICCCSI 2025).
- **Perdana, G. A.**, Wijaya, I.I., Fahreza, K.A., Tarigan, G.A. "Ablation Study: Calibrated Adaptive Learning Ensemble Methodology." *Procedia Computer Science* (ICCCSI 2025). **First author. [1 Citation]**

AWARDS & HONORS

Best Presenter Award

Semarang, Indonesia

The 10th International Conference on Computer Science (ICCCSI)

August 2025

- Awarded for technical communication and research defense among international doctoral and academic presenters.

EXPERIENCE

Apple Developer Academy Scholar

Feb. 2026 – Present

Apple & Binus University

BSD City, Indonesia

- Selected for the 2026 cohort to engineer native iOS/macOS applications using Swift, SwiftUI, and CoreML.
- Collaborating within cross-functional teams employing Challenge-Based Learning to design, build, and ship production-ready applications to the App Store.

Undergraduate Research Assistant

Jan 2025 – Dec 2025

Binus University (Bina Nusantara University)

Anggrek Campus, Jakarta

- **Lead Author & Researcher:** Led experimental design for bone fracture detection, progressing from research assistant to first authorship on an IEEE publication.
- **Data Engineering:** Designed a preprocessing pipeline to mitigate severe class imbalance in X-ray data.
- **Model Architecture:** Benchmarked FPN vs. PAFPN architectures; engineered a Faster R-CNN model with an EfficientNetV2-S backbone, implementing cost-sensitive loss functions.
- **Training Pipeline:** Optimized training using AdamW, mixed precision (AMP), and Exponential Moving Average (EMA); achieved a 21.6% mAP improvement over baseline models.

Freelance Software Developer & Independent Tutor

Aug. 2021 – Jan. 2026

Self-Employed

Remote & Jakarta, Indonesia

- Various contract work including full-stack development, research consulting, and courier/logistics services, averaging 80+ hours/week while maintaining full-time student status.
- Delivered technical projects ranging from web applications to research (Computer Vision) methodology consultation.

SELECTED PROJECTS

MESA-ViT: Multi-Scale Efficient Spatial Attention Vision Transformer	Aug. 2025 – Present
- Architected a 4-stage isotropic hybrid Vision Transformer (MESA-ViT) from scratch in PyTorch, built on a composite block (MESA → XCA+LPI → FFN) that mixes spatial and channel features in near-parallel. Paper in preparation for CVPR 2027. - Engineered a Multi-Scale Efficient Spatial Attention (MESA) module with multi-scale dilated key-value extraction that runs cheaper than standard self-attention, paired with a Cross-Covariance Attention (XCA) sub-layer for resolution-linear channel mixing. - Added a parallel ConvNeXt-style CNN stem with content-gated lateral fusion, and built the training stack on 2D RoPE, bf16 mixed precision, and NVIDIA DALI; held the Tiny variant to ~9.60M parameters.	
CALM: Calibrated Adaptive Learning via Mutual-Ensemble Fusion	Feb. 2025 – May 2025
- Designed a novel ensemble framework integrating knowledge distillation, mutual learning, and calibration to enhance model confidence and reliability. - Introduced adaptive curriculum protocol to dynamically schedule training objectives, optimizing multi-teacher fusion. - Achieved 97.16% accuracy on CIFAR-10 with a 20% reduction in Expected Calibration Error (ECE), demonstrating strong generalization.	
Diabetic Retinopathy Detection System	Sep. 2024 – Nov. 2024
- Engineered a desktop analysis tool for retinal disease classification featuring Grad-CAM visualization for medical interpretability. - Fine-tuned InceptionV3 on the APTOS dataset, using data augmentation to achieve 88% accuracy.	
Ensemble Learning for Waste Classification	Mar 2025 – May 2025
Developed a logit-fusion ensemble combining EfficientNet-B0 and DenseNet121 with learnable weights, improving classification performance on 12-class datasets.	

LEADERSHIP & SERVICE

Board Member – General Secretary, Bagi Dunia (NGO)	Feb. 2024 – Present
<i>Jakarta, Indonesia</i> - Directed financial operations and fundraising, securing Rp17M+ (USD ≈\$1000) and seven corporate sponsorships. - Managed 30+ volunteers and coordinated logistics for food distribution aid to over 100 beneficiaries. - Supervised volunteering volumes, Treasury & Public Relations Operations.	
Academic Tutor	Nov. 2023 – Dec 2025
<i>Independent</i> Conducted weekly tutorials in Statistics, Discrete Math, Computational Physics and Linear Algebra; organized technical bootcamps reaching 150+ students.	<i>Jakarta, Indonesia</i>

CERTIFICATIONS

Course Certifications
NVIDIA Deep Learning Institute (Dec 2025): Building LLM Applications With Prompt Engineering Kaggle ML (2025): Computer Vision, Model Explainability, Deep Learning FreeCodeCamp (2024-2025): Scientific Computing (Python), Machine Learning (Python), Algorithms

TECHNICAL SKILLS

Languages: Python, C, SQL, JavaScript, Java
Deep Learning: PyTorch, TensorFlow, Keras, Vision Transformers (ViT), CNNs, Object Detection, Transfer Learning
ML Ops & Tools: Git, Weights & Biases (W&B), NumPy, Pandas, OpenCV, Grad-CAM
Research: Experimental Design, Ablation Studies, Statistical Analysis, Technical Writing, Data Augmentation & Pipeline curation

LANGUAGES

English (Professional Proficiency)	Indonesian (Native)
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